

```

sanaulcse@DESKTOP-70HVUSR:~$ ./5_3perm
5 3
Enter n d (example: 5 3):
=== PAPER-style construction: (P,Q) SYT pairs -> inverse RSK -> permutations ===
n=5, d=3, radius r=d-1=2, need LIS = needFirst = 3

-----
Shape lambda = (5)
f^lambda = 1  =>  (f^lambda)^2 = 1
SYT count generated = 1
Permutations count = 1
Permutations:
1 2 3 4 5

-----
Shape lambda = (4,1)
f^lambda = 4  =>  (f^lambda)^2 = 16
SYT count generated = 4
Permutations count = 16
Permutations:
2 1 3 4 5
2 3 1 4 5
2 3 4 1 5
2 3 4 5 1
3 1 2 4 5
1 3 2 4 5
1 3 4 2 5
1 3 4 5 2
4 1 2 3 5
1 4 2 3 5
1 2 4 3 5

```

```
5 1 2 3 4
1 5 2 3 4
1 2 5 3 4
1 2 3 5 4
```

```
-----
Shape lambda = (3,2)
f^lambda = 5    =>    (f^lambda)^2 = 25
SYT count generated = 5
Permutations count = 25
Permutations:
2 1 4 3 5
2 4 1 3 5
2 1 4 5 3
2 4 1 5 3
2 4 5 1 3
3 1 4 2 5
3 4 1 2 5
3 1 4 5 2
3 4 1 5 2
3 4 5 1 2
2 1 5 3 4
2 5 1 3 4
2 1 3 5 4
2 3 1 5 4
2 3 5 1 4
3 1 5 2 4
3 5 1 2 4
3 1 2 5 4
1 3 2 5 4
1 3 5 2 4
4 1 5 2 3
4 5 1 2 3
4 1 2 5 3
1 4 2 5 3
1 4 5 2 3
```

```
Shape lambda = (3,1,1)
f^lambda = 6    =>    (f^lambda)^2 = 36
SYT count generated = 6
Permutations count = 36
Permutations:
```

```
3 2 1 4 5
3 2 4 1 5
3 4 2 1 5
3 2 4 5 1
3 4 2 5 1
3 4 5 2 1
4 2 1 3 5
4 2 3 1 5
2 4 3 1 5
4 2 3 5 1
2 4 3 5 1
2 4 5 3 1
4 3 1 2 5
4 1 3 2 5
1 4 3 2 5
4 1 3 5 2
1 4 3 5 2
1 4 5 3 2
5 2 1 3 4
5 2 3 1 4
2 5 3 1 4
5 2 3 4 1
2 5 3 4 1
2 3 5 4 1
5 3 1 2 4
5 1 3 2 4
1 5 3 2 4
5 1 3 4 2
1 5 3 4 2
1 3 5 4 2
5 4 1 2 3
5 1 4 2 3
1 5 4 2 3
5 1 2 4 3
1 5 2 4 3
```

```
1 5 2 4 3
1 2 5 4 3
```

Total permutations in ball $B(5,2) = 78$

NOTE: This is exact paper-style construction. It explodes fast for large n .

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